

LAB # 7 : Uniform, Normal, and Log Normal Distribution

Introduction

This lab is concerned with three important continuous probability distributions namely uniform, normal, and log normal distribution. You may find it convenient to print the pdf version of this lab. This lab will take one session.

Uniform Distribution

We say that a random variable X has a uniform distribution on the interval $[a, b]$, if it has density function f , given by

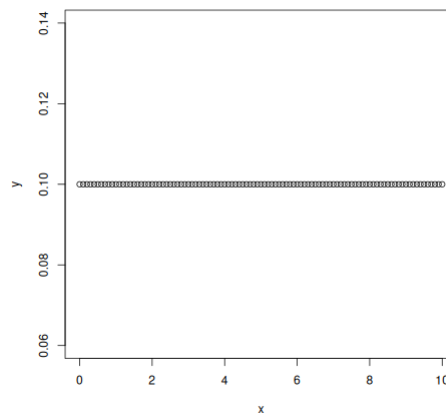
$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{elsewhere.} \end{cases}$$

We write $X \sim U[a, b]$. The random variable X can model a randomly chosen point from the interval $[a, b]$, where each choice is equally likely.

There are four basic functions in R for calculating the probability in uniform distribution.

- $dunif(x, a, b)$ returns probability density function in interval $[a, b]$. We can use this function to plot the density function.

```
x<-seq(0,10,by=0.1)
y<-dunif(x,0,10)
plot(x,y)
```



- $punif(x, a, b)$ returns the cumulative probability $P(X \leq x)$ for uniform distribution.

```
punif(5,-10,10)
[1] 0.75
punif(0:5,4,0.6)
[1] 0.50 0.55 0.60 0.65 0.70 0.75
```

- $qunif(p, a, b)$ is the inverse cdf of the uniform distribution. It returns the value x such that $punif(x, a, b) = p$.

```
qunif(c(0.25,0.5,0.75),0,10)
[1] 2.5 5 7.5
punif(2:8,0,10)
[1] 0.2 0.3 0.4 0.5 0.6 0.7 0.8
```

- $runif(n, a, b)$ generates n random values from the uniform distribution.

```
runif(10,-10,10)
[1] -2.1586878 0.7646733 6.0913344 8.8848749 -9.5967198 8.5564495
[7] -3.9578399 3.8082255 -4.7050247 -3.0242637
```

Example 1: A bus arrives every 10 minutes at a bus stop. It is assumed that the waiting time for a particular individual is a random variable with a continuous uniform distribution.

- What is the probability that the individual waits more than 7 minutes?
- What is the probability that the individual waits between 2 and 7 minutes?
- Simulate the situation by taking 100 random numbers and produce a histogram of the results.

Sol.

```
(a) 1-punif(7,0,10)
[1] 0.3
(b) punif(7,0,10)-punif(2,0,10)
[1] 0.5
(c) y <- runif(100,0,10)
hist(y)
```

Normal Distribution

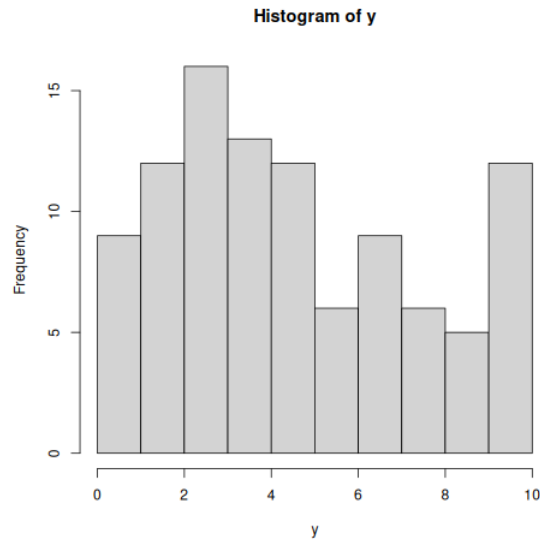
The normal (or Gaussian) distribution is the most important distribution in the study of statistics. We say that a random variable has a normal distribution with parameters μ and σ^2 if its density function f is given by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad x \in \mathbb{R}.$$

We write $X \sim N(\mu, \sigma^2)$. The parameters μ and σ^2 turn out to be the expectation and variance of the distribution, respectively. If $\mu = 0$ and $\sigma = 1$ then

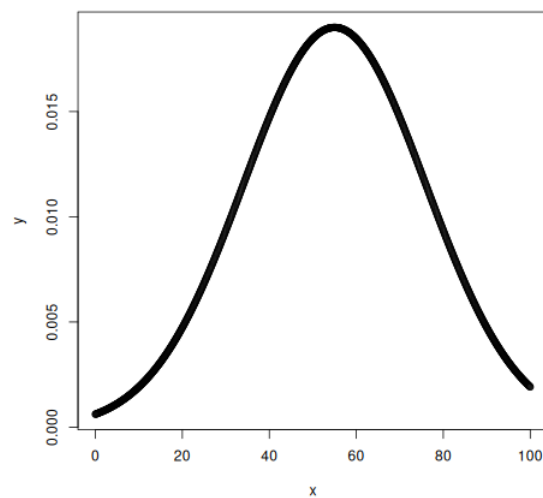
$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}},$$

and the distribution is known as a standard normal distribution. There are four basic functions in R similar to uniform distribution.



- $dnorm(x, \mu, \sigma)$ returns $P(X = x)$ for the distribution. We can use this function to plot the density function of the normal distribution.

```
x<-seq(0,100,by=0.1)
png(file="normal.png") % SAVE THE FILE as normal.png
y<- dnorm(x, 55, 21)
plot(x,y)
dev.off()
```



- $pnorm(x, \mu, \sigma)$ returns the cumulative probability $P(X \leq x)$ and x can be a vector.

```
pnorm(5, 14, 8)
[1] 0.1302945
```

```
pnorm(12:14, 14, 0.81)
[1] 0.00677199 0.10849568 0.50000000
```

- $qnorm(p, \mu, \sigma)$ returns the value x such that $pnorm(x, \mu, \sigma) = p$. This function finds the quantiles for the normal distribution.

```
qnorm(0.60, 55, 19)
[1] 59.81359
qnorm(c(0.25, 0.5, 0.75), 55, 19)
[1] 42.18469 55.00000 67.81531
```

- $rnorm(n, \mu, \sigma)$ generates n random numbers from the normal distribution.

```
rnorm(10, 55, 19)
[1] 95.30111 72.55787 16.73742 53.34533 82.81127 57.55974 42.33415
[9] 32.98337 29.07413 33.90214
```

Example: The flipper length of a certain kind of penguin are normally distributed with mean 182.5 mm and standard deviation 5.5 mm.

- What is the probability that a randomly selected penguin has a flipper less than 190 mm long? More than 190 mm?
- What is the 25th and 90th percentile for flippers length in these penguins?
- Simulate 50 random selections from this population and plot the results.

Sol.

- ```
pnorm(190, 182.5, 5.5)
[1] 0.9331928
1-pnorm(190, 182.5, 5.5)
[1] 0.0668072
```
- ```
qnorm(c(0.25, 0.90), 182.5, 5.5)
[1] 178.7903 189.5485
```
- ```
y<- rnorm(50, 182.5, 5.5)
hist(y)
```

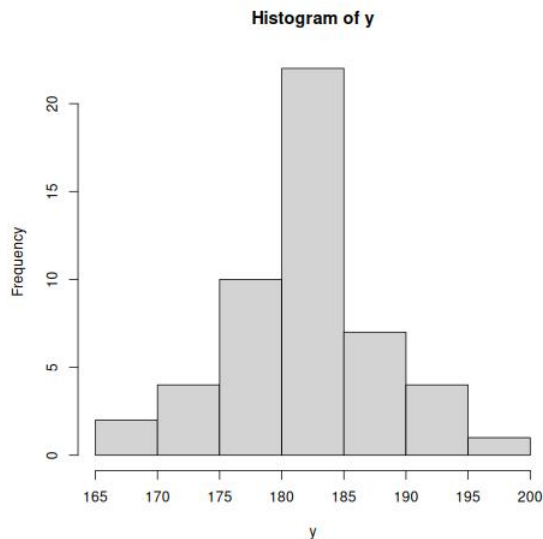
## Lognormal distribution

A continuous random variable  $X$  has a lognormal distribution if the random variable  $Y = \ln(X)$  has a normal distribution with mean  $\mu$  and standard deviation  $\sigma$ . The resulting density function of  $X$  is

$$f(x) = \begin{cases} \frac{1}{\sqrt{2\pi} \sigma x} e^{-\frac{1}{2} \left[ \frac{\ln(x) - \mu}{\sigma} \right]^2}, & x \geq 0 \\ 0, & \text{elsewhere} \end{cases}$$

There are four basic functions in R.

```
dlnorm(x, meanlog = 0, sdlog = 1, log = FALSE)
plnorm(q, meanlog = 0, sdlog = 1, lower.tail = TRUE, log.p = FALSE)
qlnorm(p, meanlog = 0, sdlog = 1, lower.tail = TRUE, log.p = FALSE)
rlnorm(n, meanlog = 0, sdlog = 1) <- seq(0:100, 55, 21)
```



**Example:** The life, in thousands of miles, of a certain type of electronic control for locomotives has an approximately lognormal distribution with  $\mu = 5.149$  and  $\sigma = 0.737$ . Find the 5th and 10th percentile of the life of such an electronic control.

Sol.

```
qlnorm(0.05, 5.149, 0.737)
[1] 51.2514
qlnorm(0.01, 5.149, 0.737)
[1] 31.0153
```

### Exercises:

- The daily amount of coffee, in liters, dispensed by a machine located in an airport lobby is a random variable  $X$  having a continuous uniform distribution with  $a = 7$  and  $b = 10$ .
  - Find the probability that on a given day the amount of coffee dispensed by this machine will be at most 8.8 liters; more than 7.4 liters but less than 9.5 liters; and at least 8.5 liters.
  - Simulate the situations by taking 100 random numbers and plot the results.
- Imagine a class's exam scores follow a normal distribution. If the average score is 60 and the standard deviation is 10.
  - What is the probability of the scores between 80 to 90? More than 90?
  - What are the scores value corresponding to first quartile and median?
  - Simulate the results for 100 random situations and plot the results.
- Suppose  $Y$  follows a lognormal distribution with parameters  $\mu = 1$  and  $\sigma = 1$ . Let  $Y_1 = 1.25Y$ .
  - Find the probability that  $Y_1$  exceed 1.
  - The 40th and 80th percentile of  $Y_1$ .